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ENGINEERING MATHEMATICS
Advanced Math 5

Determinants

Continuation ...

5. If the elements of the jth column of a determinant D are the sum $a_{ij} + b_{ij}$, then D is the sum of the determinants D' and D" in which all the columns of D, D' and D: are the same except the jth; furthermore, the jth column of D' is a_{ij}, i = 1, 2, 3,..., n, and the jth column of D" is $b_{ij,}$ I = 1, 2, 3,..., n. Similarly for rows.

6. The value of the determinant is not changed if a column is replaced by the column plus a multiple of another column. Similarly for rows.

Determinant of a 2 x 2 matrix

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$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of a 3 x 3 matrix

The determinant of a 3 x 3 matrix can be solved using the diagonal/basket method:

$$\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$\det A = \left|\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}\right|$$

det A = (aei + bfg + cdh) – (gec + hfa +idb)

Determinant of a 4 x 4 matrix

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A. Using Pivotal Element method:

Example: Find det A.

$$A = \begin{vmatrix} 2 & -4 & 3 & -1 \\ -1 & 1 & -2 & 2 \\ 3 & 2 & -4 & -1 \\ -2 & 5 & 1 & 4 \end{vmatrix}$$

$$\det A = \begin{vmatrix} 2-(-4)(-1) & 3-(-4)(-2) & -1-(-4)(2) \\ 3-(2)(-1) & -4-(2)(-2) & -1-(2)(2) \\ -2-(5)(-1) & 1-(5)(-2) & 4-(5)(2) \end{vmatrix} (1)^{2+2}$$

2 for 2nd row and 2 for 2nd column

$$\det A = \begin{vmatrix} -2 & -5 & 7 \\ 5 & 0 & -5 \\ 3 & 11 & -6 \end{vmatrix}$$

$$\det A = \left|\begin{array}{ccc|cc} -2 & -5 & 7 & -2 & -5 \\ 5 & 0 & -5 & 5 & 0 \\ 3 & 11 & -6 & 3 & 11 \end{array}\right|$$

det A = 2(0)(-6) + (-5)(-5)(3) + 7(5)(11)
– [3(0)(7) + (11)(-5)(-2)
+ (-6)(5)(-5)]

det A = 200

B. Using Modification method:

$$A = \begin{vmatrix} 2 & -4 & 3 & -1 \\ -1 & 1 & -2 & 2 \\ 3 & 2 & -4 & -1 \\ -2 & 5 & 1 & 4 \end{vmatrix}$$

Set the encircled numbers to zero by:
Multiplying column 2 by 1 and add it to column 1
Multiplying column 2 by 2 and add it to column 3
Multiplying column 2 by –2 and add it to column 4
The new matrix becomes,

$$A = \begin{vmatrix} -2 & -4 & -5 & 7 \\ 0 & 1 & 0 & 0 \\ 5 & 2 & 0 & -5 \\ 3 & 5 & 11 & -6 \end{vmatrix}$$

$$A = (1)\begin{vmatrix} -2 & -5 & 7 \\ 5 & 0 & -5 \\ 3 & 11 & -6 \end{vmatrix}(-1)^{2+2}$$

Set the encircled number to zero by multiplying column 1 by 1 and add it to column 3

$$A = \begin{vmatrix} -2 & -5 & 5 \\ 5 & 0 & 0 \\ 3 & 11 & -3 \end{vmatrix}$$

2 for 2nd row
1 for 1st column

$$\det A = (5)\begin{vmatrix} -5 & 5 \\ 11 & -3 \end{vmatrix}(-1)^{2+1}$$

= 5 [(-5)(-3) – (11)(5)] (-1)

det A = 200

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Laplace Transform

Laplace Transform, named after the French mathematician Pierre-Simon de Laplace (1749-1827), is a continuous-time signal representation in terms of complex exponentials. This type of transform can give you broader characterization of systems and their interaction with signals.

The Laplace transform of a function f(t) denoted by L [f(t)] or F(s) is defined as a function of a variable "s" by the integral:

$$F(s) = \ell\left[f(t)\right] = \int_0^\infty f(t)\, e^{-st} dt$$

where:
t > 0 and s is any number (real or complex)

Hence, the Laplace transform is an integral operation.

Laplace transform of some elementary functions:

	f(t)	F(s)
1.	1	$\frac{1}{s}$
2.	t	$\frac{1}{s^2}$
3.	t^n	
4.	$e^{\pm at}$	$\frac{1}{s \mp a}$
5.	$t^n\ e^{\pm at}$	$\frac{n!}{(s \mp a)^{n+1}}$

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